Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) MODEL EXAMINATION, 2026

THIRD SEMESTER

Skill Development Components (SDC)

MG3SDCSNP200 – BIOMECHANICS

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Short Answer Questions

Answer any 10 questions, Each question carries 2 marks

1. Define biomechanics. Mention any two areas of its importance in physiotherapy. [K] / [CO1]
2. Differentiate between linear motion and angular motion. [U] / [CO1]
3. What is mechanical advantage? Mention its significance in human movement. [U] / [CO1]
4. What is a closed kinetic chain? Give one example from the lower limb. [K] / [CO1]
5. List the three structural classifications of joints. [K] / [CO2]
6. Mention any four structural components of a synovial joint. [K] / [CO2]
7. What are the main functions of the rib cage? [U] / [CO2]
8. Name the major regions of the vertebral column. [K] / [CO2]
9. Mention the major movements possible at the temporomandibular joint (TMJ). [K] / [CO2]
10. What is scapulohumeral rhythm? [K] / [CO3]
11. Define prehension and mention its importance in hand function. [K] / [CO3]
12. Define gait cycle. [K] / [CO4]

[2 × 10 = 20 Marks]

PART B

Short Essay Questions

Answer any 6 questions, Each question carries 5 marks

13. Explain the planes and axes of human movement with suitable examples. [U] / [CO1]

14. Describe the three types of levers in the human body with suitable examples. [U] / [CO1]

15. Explain the structure and functional characteristics of synovial joints, including degrees of freedom. [U] / [CO2]

16. Describe the biomechanics of the thorax and chest wall during ventilatory movements. [U] / [CO2]

17. Explain the regional biomechanics of the vertebral column, with reference to the cervical, thoracic and lumbar regions. [U] / [CO2]

18. Discuss the functional biomechanics of the shoulder complex. [An] / [CO3]

19. Explain the structural and functional biomechanics of the knee and ankle-foot complexes. [An] / [CO3]

20. Describe the normal gait cycle and its major phases. [U] / [CO4]

21. Explain common gait malalignments and their biomechanical consequences. [An] / [CO4]

[5 × 6 = 30 Marks]

PART C

Essay Questions

Answer any 2 questions, Each question carries 10 marks

22. Explain the basic biomechanical concepts of force, work, energy, power, equilibrium and friction, and discuss their application to human movement. [U] / [CO1]

23. Discuss in detail the biomechanics of the vertebral column, including regional structure and function and the role of the muscles of the vertebral column. [An] / [CO2]

24. Analyse the functional biomechanics of the hip, knee, ankle and foot during weight-bearing and common functional activities. [An] / [CO3]

25. Explain the biomechanics of normal and pathological gait, and discuss the biomechanical principles of posture and ergonomics in physiotherapy practice. [E] / [CO4]

[10 × 2 = 20 Marks]